US 20250270825A1

(19) United States
(12) Patent Application Publication
Labonte et al.
(10) Pub. No.: US 2025/0270825 A1
(43) Pub. Date: Aug. 28, 2025

(54) **CONNECTOR FOR TRANSITION MOLDINGS**

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(21) Appl. No.: **18/952,058**

(22) Filed: **Nov. 19, 2024**

(30) **Foreign Application Priority Data**

Feb. 27, 2024 (GB) 2402794.8

Publication Classification

(51) **Int. Cl.**
E04F 19/06 (2006.01)

(52) **U.S. Cl.**
CPC ***E04F 19/066*** (2013.01)

(57) **ABSTRACT**

The invention provides a flooring molding connector for transitional molding. The system features a base member that secures to a sub-floor through adhesive or mechanical means, and a serrated insertion member for engaging a complementary molding member. The molding member is equipped with a flat section and a downward-extending stem that ends in an arrowhead tip, facilitating secure engagement with the insertion member.

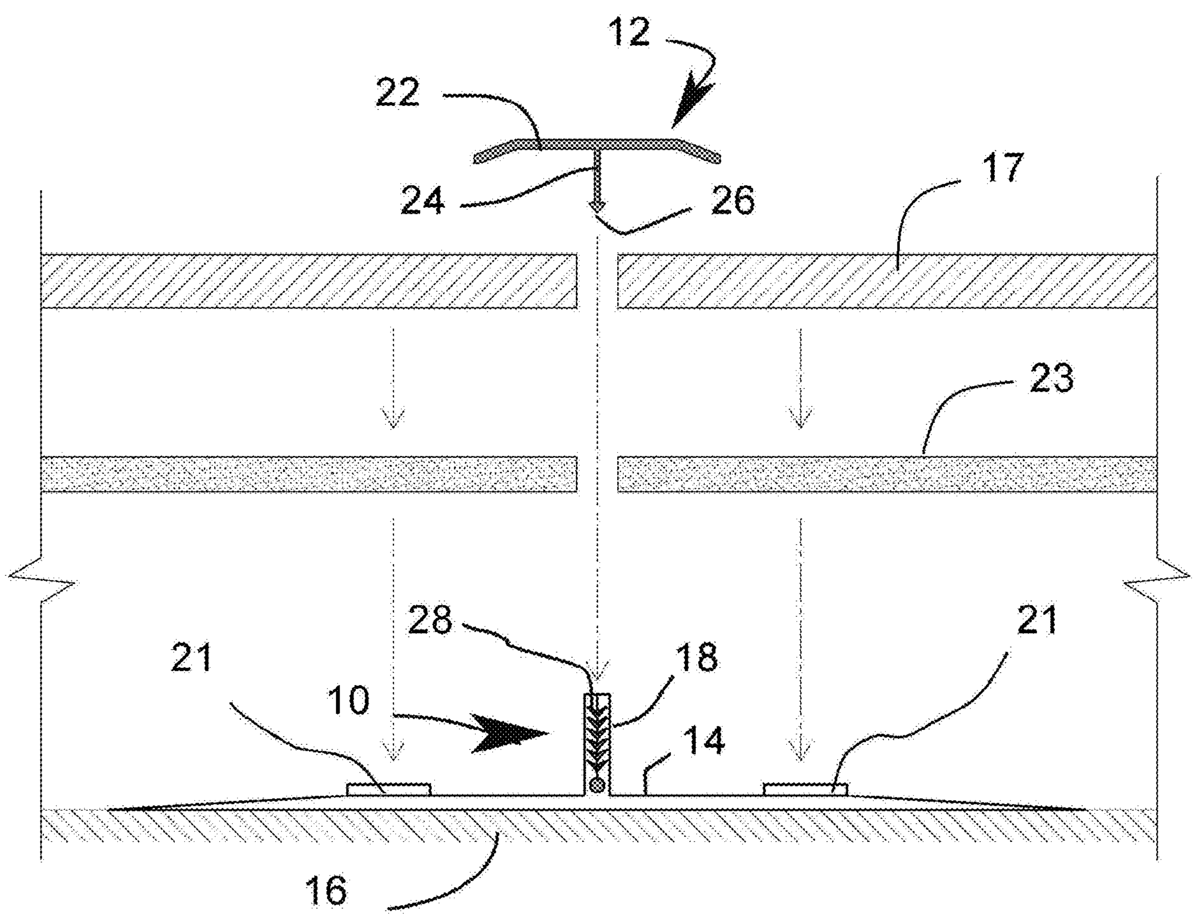


FIG. 1

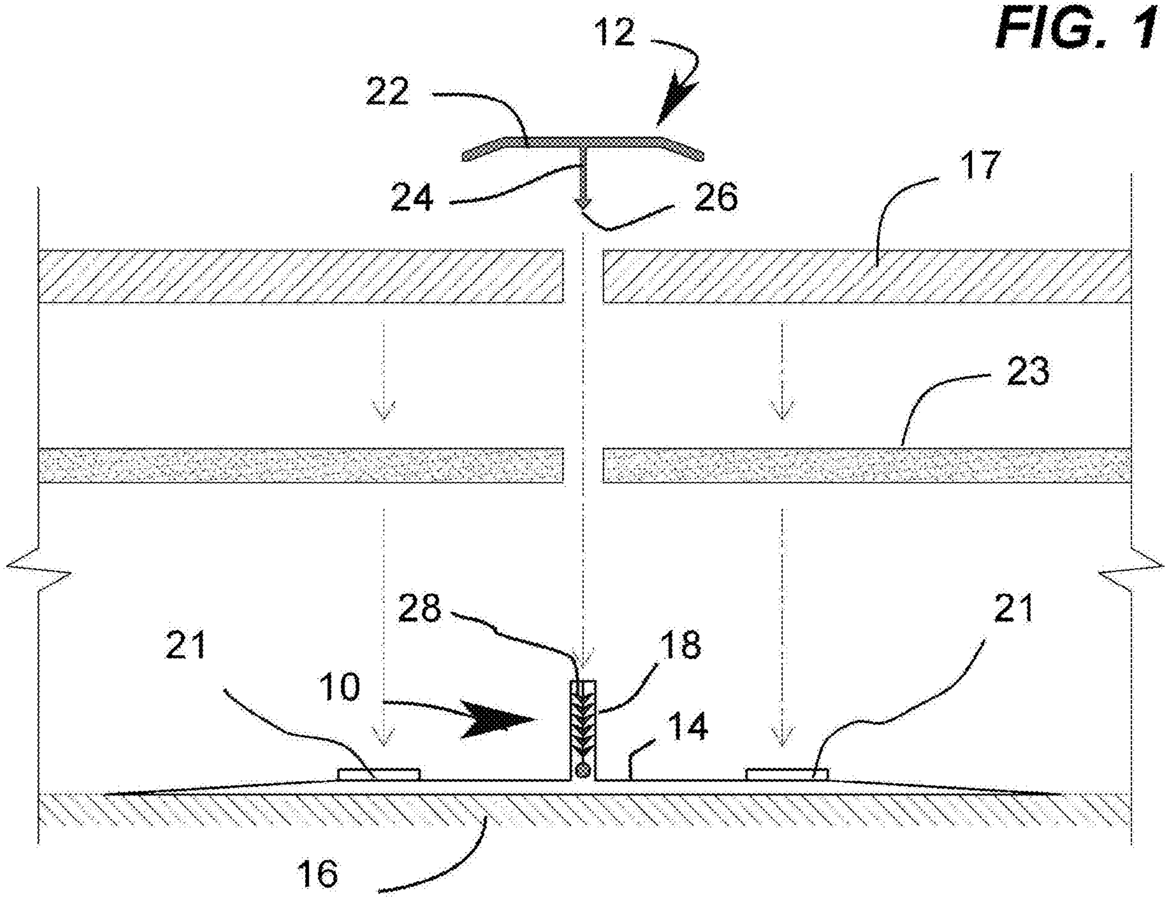


FIG. 2

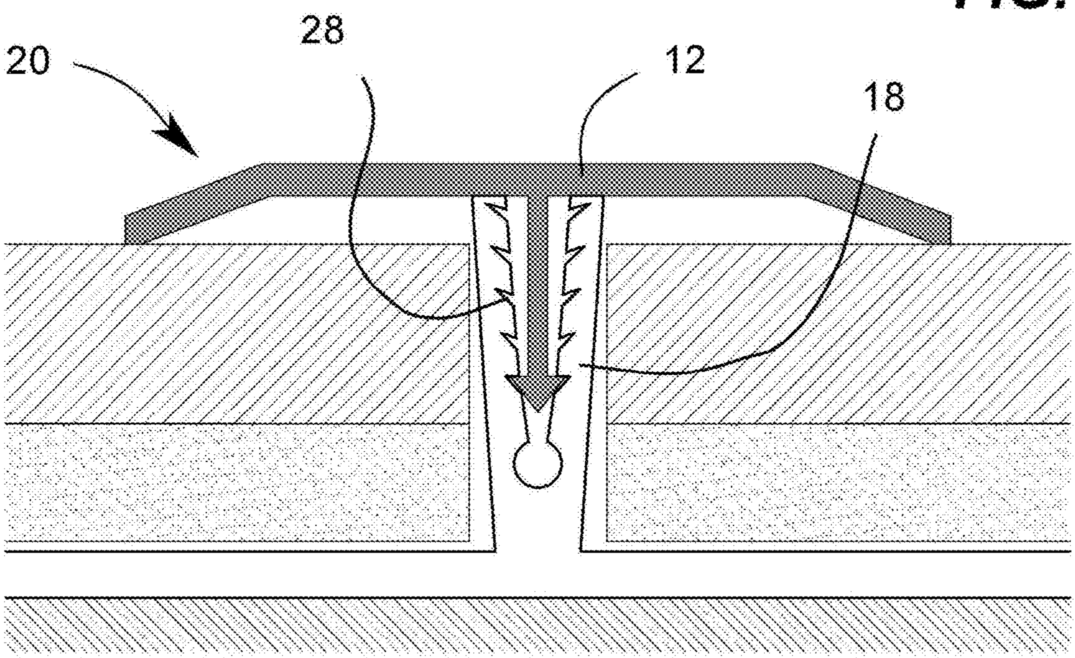
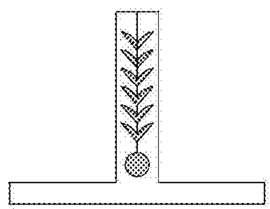


FIG. 3



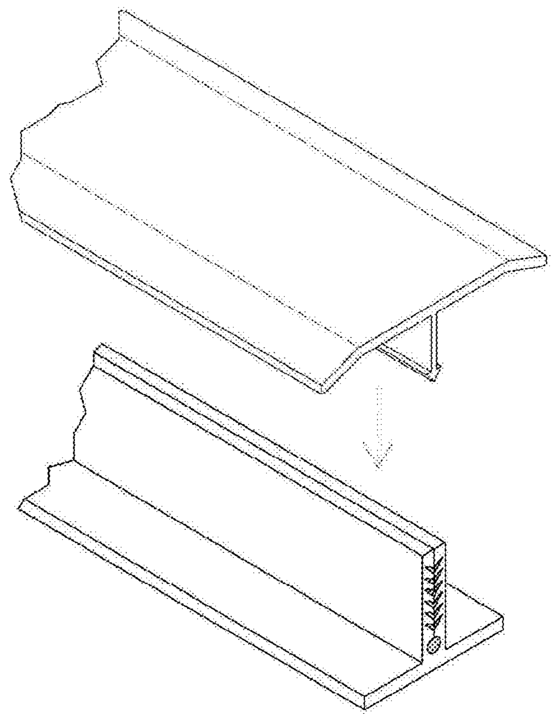


FIG. 4

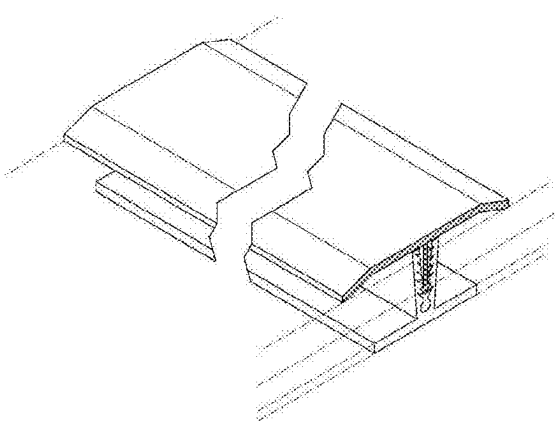


FIG. 5

CONNECTOR FOR TRANSITION MOLDINGS

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] The present application claims priority to application number GB2402794.8 filed on Feb. 27, 2024, the disclosure of which is hereby incorporated in its entirety at least by reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

[0002] The present invention relates generally to flooring hardware but more particularly to a connector for transition moldings.

2. Description of Related Art

[0003] The utilization of moldings to interface between different sections of flooring has been established for an extended period. Numerous models of moldings exist, each differentiated by aesthetic aspects such as color and shape. However, the method of attachment to the subfloor remains consistent across these variations. Specifically, while the visible part of the molding can vary in appearance, the portion concealed beneath the flooring material and affixed to the subfloor is uniform in structure and function.

BRIEF SUMMARY OF THE INVENTION

[0004] The following presents a simplified summary of some embodiments of the invention in order to provide a basic understanding of the invention. This summary is not an extensive overview of the invention. It is not intended to identify key/critical elements of the invention or to delineate the scope of the invention. Its sole purpose is to present some embodiments of the invention in a simplified form as a prelude to the more detailed description that is presented later.

[0005] An object of the present invention is to provide a connector system for transitional molding that facilitates easy and secure installation onto a sub-floor. This system aims to enhance the convenience and reliability of installation processes, while offering adjustable engagement features to accommodate various molding sizes and profiles, thereby improving both aesthetic integration and structural stability in flooring applications.

[0006] In order to do so, a connector system for securing transitional molding to a sub-floor is provided, comprising a base member configured to be fastened onto the sub-floor; an insertion member arising perpendicularly from the base member, wherein the insertion member comprises a plurality of serrations; and a molding member including a flat section and a stem extending perpendicularly and downwardly from the flat section, the stem terminating in an arrowhead tip configured for engagement with the plurality of serrations of the insertion member, wherein the engagement permits one-way insertion.

[0007] In one embodiment, the base member is attachable to the sub-floor via adhesive or mechanical fastening. In one embodiment, the plurality of serrations on the insertion member are designed to engage with the arrowhead tip of the stem member to secure the molding member in place and prevent unintentional dislodgement.

[0008] In another aspect of the invention, a method for installing a transitional molding onto a sub-floor is provided, comprising providing a connector having a base member configured to be fastened to the sub-floor, an insertion member arising perpendicularly from the base member having a plurality of serrations; providing a molding member including a flat section with a stem extending perpendicularly and downwardly terminating in an arrowhead tip; aligning the arrowhead tip of the molding member with the plurality of serrations of the insertion member; and inserting the arrowhead tip into the insertion member to secure the molding member in place, wherein the engagement of the arrowhead tip with the plurality of serrations permits one-way insertion.

[0009] In one embodiment, the base member is fastened to the sub-floor using a selection between adhesive and mechanical fastening based on sub-floor material or installer preference.

[0010] The foregoing has outlined rather broadly the more pertinent and important features of the present disclosure so that the detailed description of the invention that follows may be better understood and so that the present contribution to the art can be more fully appreciated. Additional features of the invention, which will be described hereinafter, form the subject of the claims of the invention. It should be appreciated by those skilled in the art that the conception and the disclosed specific methods and structures may be readily utilized as a basis for modifying or designing other structures for carrying out the same purposes of the present disclosure. It should be realized by those skilled in the art that such equivalent structures do not depart from the spirit and scope of the invention as set forth in the appended claims.

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] Other features and advantages of the present invention will become apparent when the following detailed description is read in conjunction with the accompanying drawings, in which:

[0012] FIG. **1** is an exploded cutaway side view of the connector for transition molding according to an embodiment of the present invention.

[0013] FIG. **2** is a cutaway side view of the connector for transition molding installed according to an embodiment of the present invention.

[0014] FIG. **3** is a cutaway side view of the base member according to an embodiment of the present invention.

[0015] FIG. **4** is an isometric view of connector for transition molding prior to the molding inserted within the insertion member according to an embodiment of the present invention.

[0016] FIG. **5** is an isometric view of molding installed on the connector for transition molding according to an embodiment of the present invention.

DETAILED DESCRIPTION OF THE INVENTION

[0017] The following description is provided to enable any person skilled in the art to make and use the invention and sets forth the best modes contemplated by the inventor of carrying out his invention. Various modifications, however, will remain readily apparent to those skilled in the art, since the general principles of the present invention have been defined herein to specifically provide a connector for transition molding.

[0018] It is to be understood that the terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting. The terms “a” or “an,” as used herein, are defined as to mean “at least one.” The term “plurality,” as used herein, is defined as two or more. The term “another,” as used herein, is defined as at least a second or more. The terms “including” and/or “having,” as used herein, are defined as comprising (i.e., open language). The term “providing” is defined herein in its broadest sense, e.g., bringing/coming into physical existence, making available, and/or supplying to someone or something, in whole or in multiple parts at once or over a period of time. These terms generally refer to a range of numbers that one of skill in the art would consider equivalent to the recited values (i.e., having the same function or result). In many instances these terms may include numbers that are rounded to the nearest significant figure.

[0019] Referring now to any of the accompanying FIGS. **1**-**5**, the present invention includes a connector **10** for transitional molding **12**. In one embodiment, the connector **10** features a base member **14** that can be either adhesively by way of adhesive strips (not shown) or mechanically fastened onto a sub-floor **16**. The choice between adhesive and mechanical fastening may depend on the material of the sub-floor or the installation preferences. Between the sub-floor **16** and the floor material **17** is an optionally installed membrane **23**. The membrane **23** is cut at an insertion member **18** rising perpendicularly from the base member **14**. The membrane is adhesively connected to the base member by way of adhesive strips **21**. The insertion member **18** is equipped with a plurality of serrations **28**.

[0020] As best seen in FIG. **2**, the serrations **28** on the insertion member **18** are designed to engage with the molding member **12**. FIG. **1** shows one embodiment where the molding member **12** includes a flat section **22** and a stem member **24** that extends perpendicularly and downwardly, terminating in an arrowhead tip **26**. This tip is specifically configured and sized for insertion into the insertion member **18**. The interaction between the serrations **28** and the arrowhead tip **26** allows for one-way insertion, creating a secure fit that prevents unintentional dislodgement.

[0021] In one embodiment, the construction of the base member **14**, insertion member **18**, and molding member **20** can utilize various materials to meet different environmental and aesthetic requirements. Suitable materials include, but are not limited to, polymers, metals, composites, or any combination of these. The selection of materials can be influenced by factors such as durability, cost, aesthetic considerations, and compatibility with the flooring material. Further, the dimensions of both the connector and the molding members are adaptable to suit different flooring and molding designs.

[0022] It is also important to note that the angle or depth of the serrations can be varied to customize the engagement strength. This flexibility facilitates not only easier installations and secure fastening but also allows for straightforward disassembly when needed.

[0023] In addition to the functional aspects, aesthetic customizations are offered to enhance the visual appeal of the installation. The visible parts of the molding member can be finished in a variety of colors and textures, designed to either match or contrast with the existing flooring materials, thereby contributing to the overall interior design. This comprehensive approach ensures that the invention is both practical and visually appealing, addressing the diverse needs of modern flooring installations.

[0024] Although the invention has been described in considerable detail in language specific to structural features, it is to be understood that the invention defined in the appended claims is not necessarily limited to the specific features described. Rather, the specific features are disclosed as exemplary preferred forms of implementing the claimed invention. In other words, the terminology and phraseology used in this description and the abstract are for illustrative purposes and should not be considered limiting. Therefore, while exemplary illustrative embodiments of the invention have been described, numerous variations and alternative embodiments will occur to those skilled in the art. Such variations and alternative embodiments are contemplated, and can be made without departing from the spirit and scope of the invention.

[0025] It should further be noted that throughout the entire disclosure, the labels such as left, right, front, back, top, bottom, forward, reverse, clockwise, counterclockwise, up, down, or other similar terms such as upper, lower, aft, fore, vertical, horizontal, oblique, proximal, distal, parallel, perpendicular, transverse, longitudinal, etc. have been used for convenience purposes only and are not intended to imply any particular fixed direction or orientation. Instead, they are used to reflect relative locations and/or directions/orientations between various portions of an object.

[0026] In addition, references to “first,” “second,” “third,” and so fourth members throughout the disclosure (and in particular, claims) are not used to show a serial or numerical limitation but instead are used to distinguish or identify the various members of the group.

What is claimed is:

1. A connector system for securing transitional molding to a sub-floor, comprising:

a base member configured to be fastened onto the sub-floor;

an insertion member arising perpendicularly from the base member, wherein the insertion member comprises a plurality of serrations; and

a molding member including a flat section and a stem extending perpendicularly and downwardly from the flat section, the stem terminating in an arrowhead tip configured for engagement with the plurality of serrations of the insertion member, wherein the engagement permits one-way insertion.

2. The connector system of claim **1**, wherein the base member is attachable to the sub-floor via adhesive or mechanical fastening.

3. The connector system of claim **1**, wherein the plurality of serrations on the insertion member are designed to engage with the arrowhead tip of the stem member to secure the molding member in place and prevent unintentional dislodgement.

4. A method for installing a transitional molding onto a sub-floor, comprising:

providing a connector having a base member configured to be fastened to the sub-floor, an insertion member arising perpendicularly from the base member having a plurality of serrations;

providing a molding member including a flat section with a stem extending perpendicularly and downwardly terminating in an arrowhead tip;

aligning the arrowhead tip of the molding member with the plurality of serrations of the insertion member; and

inserting the arrowhead tip into the insertion member to secure the molding member in place, wherein the engagement of the arrowhead tip with the plurality of serrations permits one-way insertion.

5. The method of claim **4**, wherein the base member is fastened to the sub-floor using a selection between adhesive and mechanical fastening based on sub-floor material or installer preference.

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